DRIVEABILITY—BI-FUEL CNG—NO START IN CNG MODE WITH ALT FUEL GREEN LIGHT ON	Article No. 02-10-3

FORD: 2001-2002 F-150

ISSUE

Some 2001-2002 Bi-Fuel CNG F-150 vehicles may exhibit a crank no start condition while operating in Alternate fuel mode, with the Alt Fuel Light on. (Vehicle will start in Gasoline Mode). This may be caused by a contaminated Compuvalve, contaminated or faulty Low Flow Injectors or a faulty Compuvalve system board.

ACTION

Refer to the following Service Procedure to determine diagnostic requirements for no start condition in CNG with ALT fuel green light on.

SERVICE PROCEDURE

Refer to the following Service Procedure flow charts (Figures 1 and 2) to determine diagnostic requirements for no start condition in CNG with ALT fuel green light on.

PART NUMBER	PART NAME
1L3Z-9F954-BA	Compuvalve - 2001 Bi-Fuel CNG F-150
2L3Z-9F954-BB	Compuvalve - 2002 Bi-Fuel CNG F-150

OTHER APPLICABLE ARTICLES: 01-22-1
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 206000, 602300, 603300, 608000, 608400, 611000, 611500, 612000, 612500, 614000, 614500, 690000

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TEST STEP	RESULT	ACTION TO TAKE
No Start—CNG (Starts in Gasoline) (Fuel Selector to ALT—Light on Steady)		
NOTE: Before proceeding with the diagnostics, ensure that there are no codes displayed in the PCM or the AFCM, verify the engine operations in the gasoline fuel mode and that the CNG fuel tank is full.		
A. Connect NGS and Identify Vehicle. 1. With engine running in gasoline, monitor MAP and RPM from the AFCM, **NOTE:** Vary engine speeds and load when monitoring. **NOTE:** For the F-150, it may be necessary to ID the vehicle as an F-250 to access the AFCM. DID THE **RPM** AND **MAP** REFLECT ACTUAL OPERATING CONDITIONS?	YES → NO →	Go to **Step B**. Follow normal diagnostic procedures.
B. Check Voltage. 1a. With the fuel selector in the ALT position, disconnect connector marked J/P14 (C4014 for 2001 and C1289 for 2002). b. With key On, measure the voltage from the male connector pin D to ground. IS THE VOLTAGE MEASURED EQUAL TO THE BATTERY VOLTAGE?	YES → NO →	Go to **Step C**. Repair wiring and verify the repair.
C. Check the Control Circuit. 1a. With the connector marked J/P14 (C4014 for 2001 and C1289 for 2002) disconnected. b. Connect test light (noid light) across pin A and D of the male connector marked J/P14. c. Crank the engine while observing the light. DOES THE LIGHT COME ON AND FLASH?	YES → NO →	Go to **Step E**. Go to **Step D**.

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Figure 1 - Article 02-10-3

TEST STEP	RESULT	ACTION TO TAKE
D. Check the Main Harness Resistance. **CAUTION: Always disconnect the red power connection marked J/P26 before connecting or disconnecting the main compuvalve harness marked J25. Failure to disconnect power could result in electronic damage to the compuvalve.** **CAUTION: When pinning out the connector marked J25 connector, use a small pin (gently) to prevent damage to the connector.** 1a. With the Key Off and connector marked J/P14 (C4014 f0r 2001 and C1289 for 2002) disconnected, disconnect connector marked J25 (C4025 for 2001 and C1294 for 2002). b. Measure the resistance between connector marked J25 pin 17 and the male connector marked J/P14 pin A. IS THE RESISTANCE LESS THAN 5 OHMS?	YES → NO →	Replace the compuvalve, reconnect all connectors and verify the repair. Replace the main harness, reconnect all connectors and verify the repair.
E. Open Injector by Applying Battery Voltage. 1a. Using a jumper, apply Batt. + to connector marked J/P14 (C4013 for 2001and C1289 for 2002) female connector pin D. b. Using a jumper, apply ground to connector J/P14 (C4014 for 2001 and C1289 for 2002) female connector pin A. c. Apply battery voltage for no more than 5 seconds. **NOTE:** When the injectors open, there will be an audible click, then the sound of fuel flow. **NOTE:** If no click is heard, disconnect the battery voltage for 5 seconds, then re-apply. DOES THE INJECTOR OPEN AND FUEL FLOW?	YES → NO →	Go to **Step F**. Replace low flow injector assembly, reconnect all connectors and verify the repair.
F. Check Vehicle Operation 1a. Reconnect all connectors and place the fuel selector switch in the ALT position. b. Start the engine. DOES THE ENGINE START?	YES → NO →	Follow the CNG flush procedures to flush the injectors and verify the repair. Replace the compuvalve and verify the repair.

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Figure 2 - Article 02-10-3